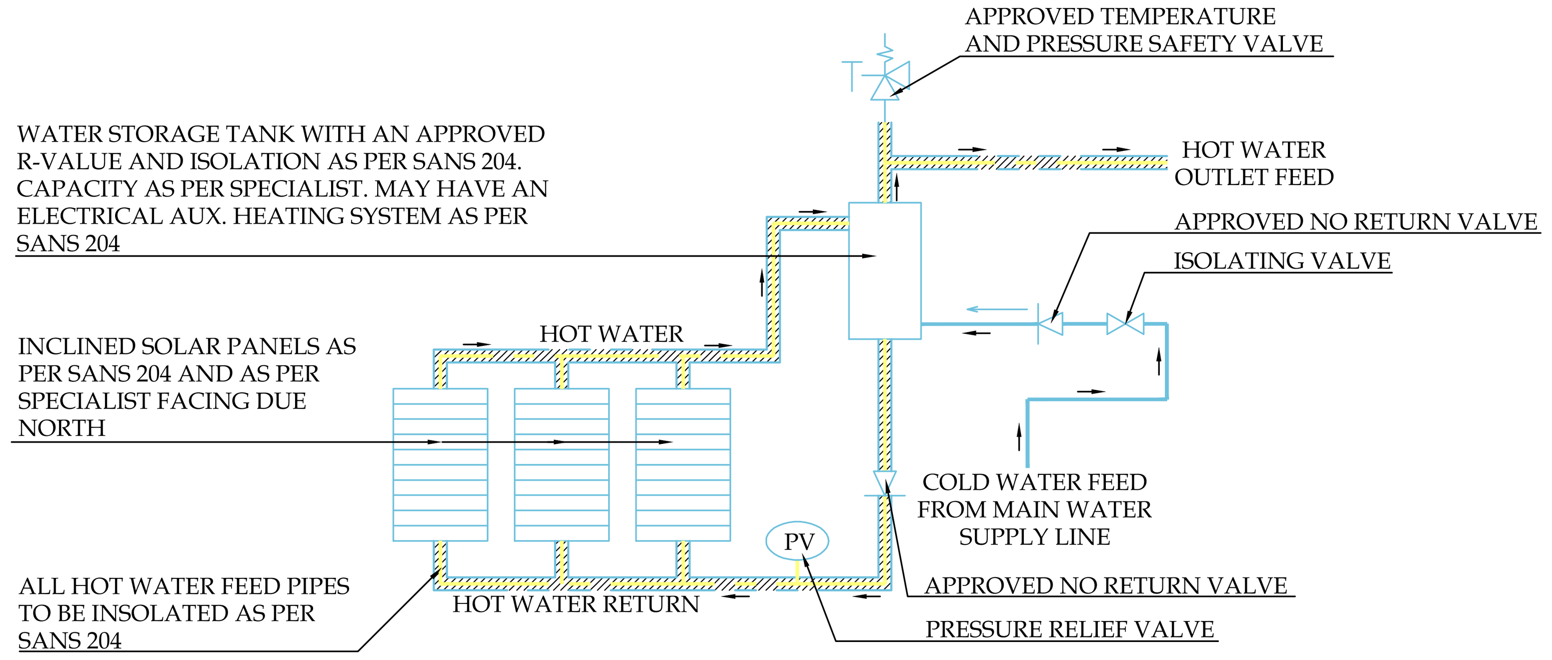


THERMOSIPHONIC HOT WATER SYSTEM DETAIL

SCALE N.T.S

ECO FLEX INSULATION TO HOT WATER PIPES TO SANS 10177-9 AND SANS 1077-10

ALL HOT WATER PIPES TO BE INSULATED TO ACHIEVE AN R VALUE OF 1

HOT WATER SYSTEM BE BE MAINTAINED IN ACCORDANCE WITH SANS 10252-1

SOLAR WATER HEATING SYSTEM TO SANS 1307 AND 10106. THERMAL PERFORMANCE TO BE DETERMINED IN ACCORDANCE WITH SANS 6211-1 AND 6211-2 INSTALLATION THEREOF WILL BE IN ACCORDANCE WITH SANS 10254

HOT WATER VESSELS AND TANKS TO HAVE A MIN R VALUE OF 2

PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1M OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITH IN 1M OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM. WHERE POSSIBLE, LENGTH OF PIPE RUNS SHOULD BE MINIMIZED

50% of HOT WATER TO COME FROM ALTERNATIVE SOURCE.
GEYSER TO BE MAINTAINED AT 60° C

SPECIFICATION OF WATERPIPING MATERIAL:
PIPES > 75mm
CLASS 12 U PVC PIPES TO SABS 966 WITH FLEXIBLE JOINTS.
PIPES < 75mm
EITHER HDPE TYPE IV CLASS 12 TO SABS 533 WITH PLASSON JOINTS OR EQUAL (SUCH AS POLY
PROPYLENE PIPE).

HOT WATER SERVICES

A MINIMUM OF 50 % BY VOLUME OF THE ANNUAL AVERAGE HOT WATER HEATING REQUIREMENT SHALL BE PROVIDED BY MEANS OTHER THAN ELECTRICAL RESISTANCE HEATING, INCLUDING, BUT NOT LIMITED TO, SOLAR HEATING, HEAT PUMPS, HEAT RECOVERY FROM OTHER SYSTEMS OR PROCESSES ETC.

THE SOLAR WATER HEATING SYSTEMS SHALL COMPLY WITH SANS 1307 AND SANS 10106, BASED ON THE THERMAL PERFORMANCE DETERMINED IN ACCORDANCE WITH THE PROVISIONS OF SANS 6211-1 AND SANS 6211-2. THE INSTALLATION THEREOF SHALL COMPLY WITH SANS 10254.

HOT WATER USAGE SHOULD BE MINIMIZED AND THE SYSTEM MAINTAINED IN ACCORDANCE WITH THE REQUIREMENTS GIVEN IN SANS 10252-1.

ALL EXPOSED PIPES TO AND FROM THE HOT WATER CYLINDERS AND CENTRAL HEATING SYSTEMS SHALL BE INSULATED WITH PIPE INSULATION MATERIAL WITH AN R-VALUE AS FOLLOW:
A PIPE WITH AN INTERNAL DIAMETER SMALLER THAN 80mm = R-VALUE OF 1.00
A PIPE WITH AN INTERNAL DIAMETER LARGER THAN 80mm = R-VALUE OF 1.50

INSULATION SHALL BE PROTECTED AGAINST THE EFFECTS OF WEATHER AND SUNLIGHT,
BE ABLE TO WITHSTAND THE TEMPERATURES WITHIN THE PIPING AND ACHIEVE THE MINIMUM TOTAL R-VALUE GIVEN ABOVE.

HOT WATER VESSELS AND TANKS SHALL BE INSULATED WITH A MATERIAL ACHIEVING A MINIMUM R-VALUE OF 2,0. NOTE TO ACHIEVE THIS VALUE, INSULATION IN ADDITION TO THE MANUFACTURERS' INSTALLED INSULATION MAY BE REQUIRED.

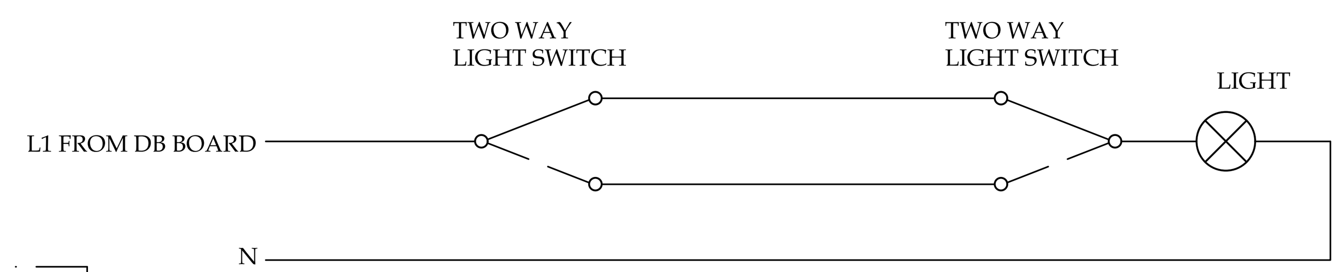
INSULATION ON VESSELS, TANKS AND PIPING CONTAINING COOLING WATER SHALL BE PROTECTED BY A VAPOR BARRIER ON THE OUTSIDE OF THE INSULATION.

THE PIPING INSULATION REQUIREMENTS DO NOT APPLY TO SPACE HEATING WATER PIPING LOCATED WITHIN THE SPACE BEING HEATED WHERE THE PIPING IS TO PROVIDE THE HEATING TO THAT SPACE OR ENCASED WITHIN A CONCRETE FLOOR SLAB OR IN MASONRY. THESE PIPES SHALL COMPLY WITH SANS 10252-1.

PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM. WHERE POSSIBLE, LENGTHS OF PIPE RUNS SHOULD BE MINIMIZED.

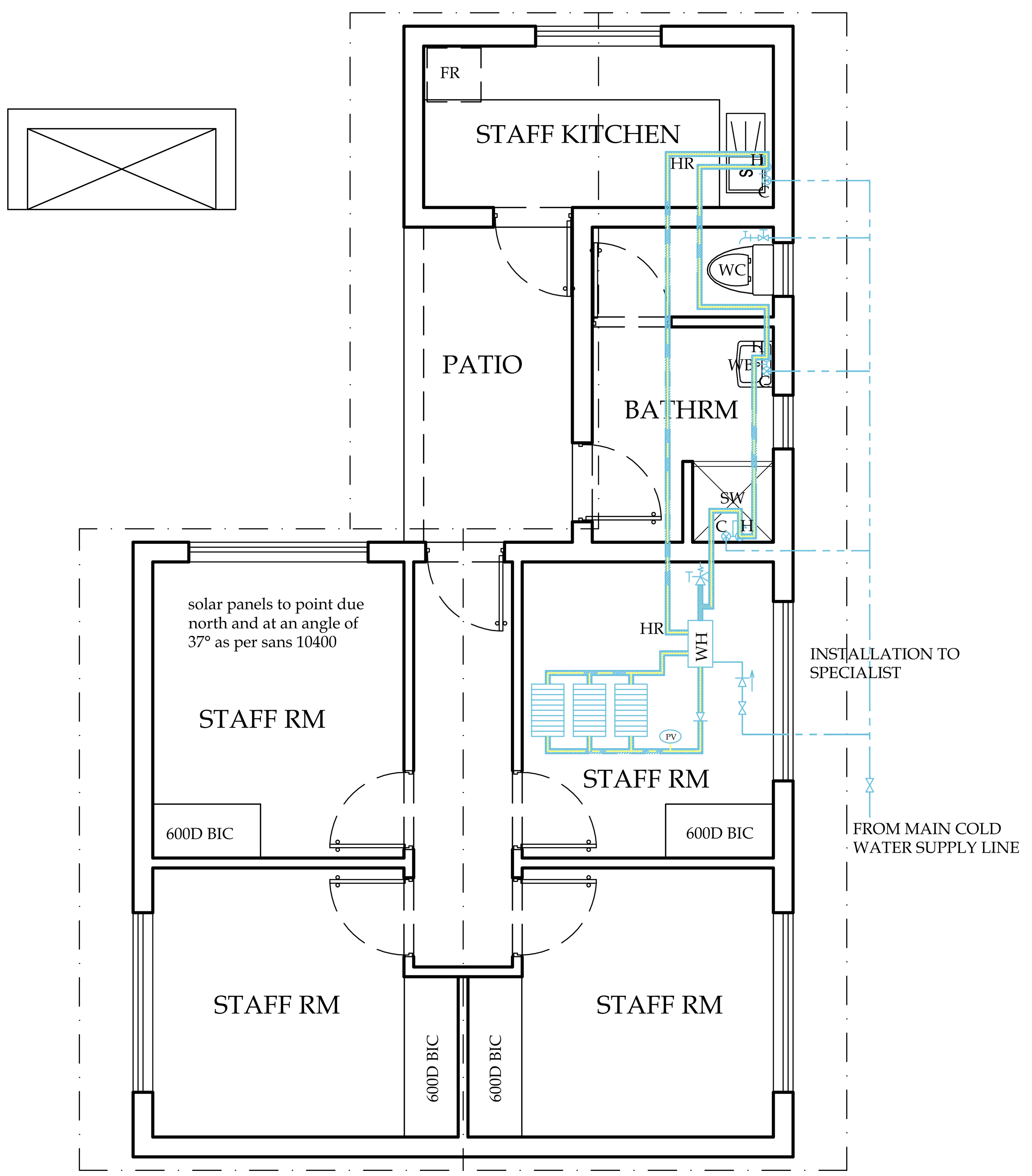
ELECTRICAL LEGEND

- SURFACE MOUNTED CEILING LIGHT LOW WATTAGE (9 OF) -5 WATT
- RECESSED BULKHEAD LIGHT LOW WATTAGE
- RECESSED INCANDESCENT CEILING LIGHT LOW WATTAGE
- RECESSED LOW-WATTAGE LED DOWNLIGHTER
- RECESSED LOW-WATTAGE WALL WASHER
- WALL MOUNTED LIGHT POINT -LED LOW WATTAGE
- WATERPROOF WALL MOUNTED FLUORESCENT BULB (1 OF)- 7 WATT
- SURFACE MOUNTED FLUORESCENT LIGHT
- RECESSED FLUORESCENT LIGHT
- LIGHTING TRACK LENGTH AS PER PLAN
- WATERPROOF SPIKE LIGHT ON ARMOUR CABLE
- LIGHT SWITCH
- WATERTIGHT LIGHT SWITCH
- DIMMER LIGHT SWITCH
- 2 WAY LIGHT SWITCH
- 15 AMP PLUG AT 220mm ABOVE F.F.L OR AS INDICATED
- DOUBLE 15 AMP PLUG AT 220mm ABOVE F.F.L OR AS INDICATED (13 OF)
- WATER PROOF DOUBLE 15 AMP PLUG AT 220mm OR AS INDICATED
- SHAVER SOCKET / HAIR DRYER OUTLET
- TELEVISION POINT AT 220mm ABOVE F.F.L OR AS INDICATED
- GEYSER CONNECTION POINT CONNECTED TO SOLAR PANELS/ HEAT PUMP. GEYSER TO HAVE A SOLAR BLANKET.
- TELEPHONE POINT AT 220mm ABOVE F.F.L OR AS INDICATED
- INTERCOM POINT AT 220mm ABOVE F.F.L OR AS INDICATED
- STOVE / OVEN CONNECTION POINT
- STOVE / OVEN ISOLATOR SWITCH
- BELL PUSH
- BELL POINT
- UNDERFLOOR HEATING THERMOSTAT
- BATHROOM HEATER POINT
- CEILING MOUNTED FAN
- OUTLET FOR ELECTRICAL POINT AS SPECIFIED ON PLAN
- DISTRIBUTION BOARD- TO SANS 10142-1
- CHANDELIER LOW WATTAGE
- EXTRACTOR FAN CONNECTED TO LIGHT SWITCH. AIR MAY NOT BE RECIRCULATED IN BATHROOMS & DRESSER
- FLUORESCENT TUBELIGHT
- WI-FI POINT



TWO WAY LIGHT SWITCHING SCHEMATIC DIAGRAM

SCALE N.T.S

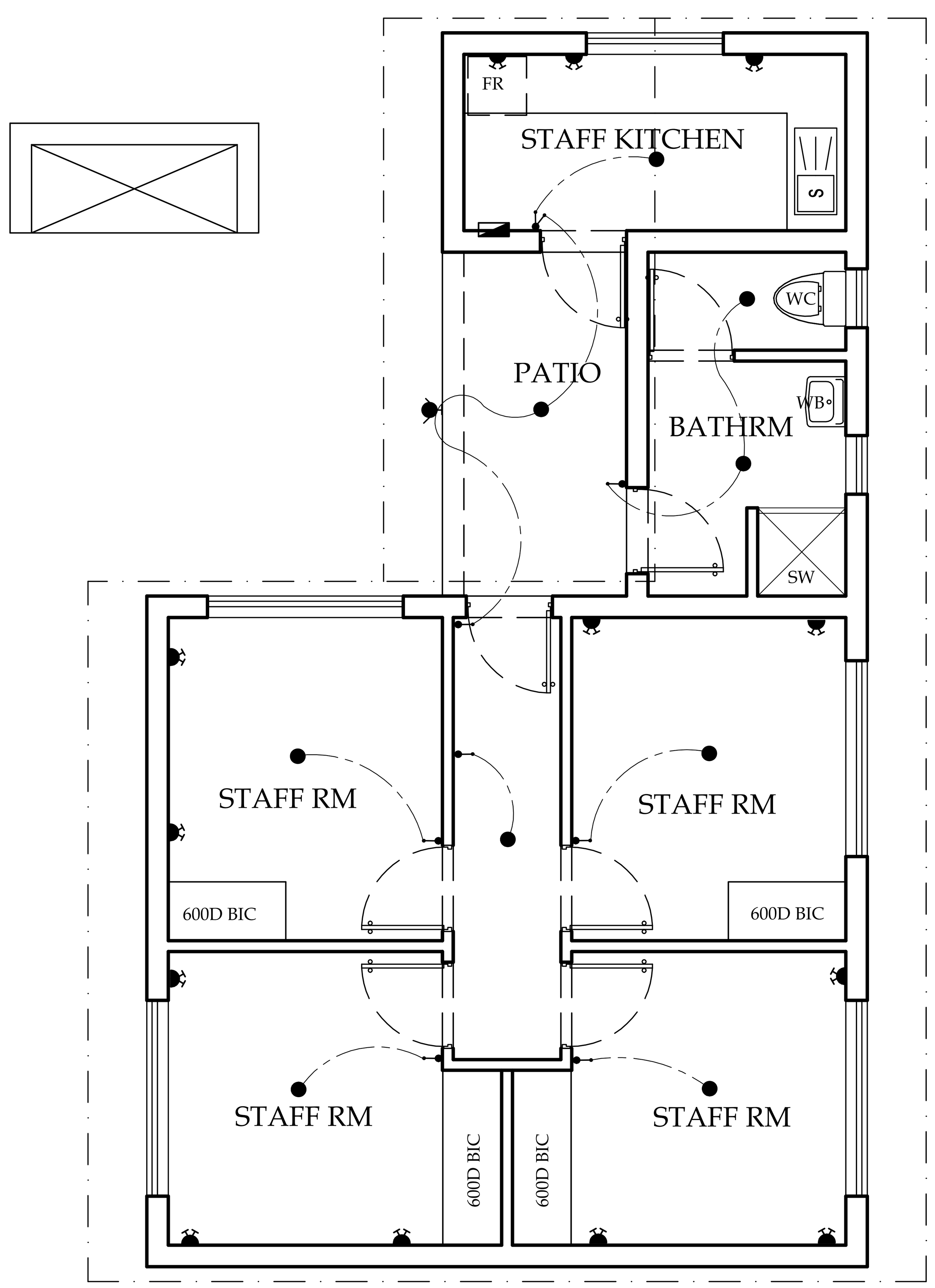


WATERPIPES SHALL HAVE A MINIMUM COVER OF 500MM
PROVISION SHOULD BE MADE FOR FUTURE CONNECTIONS IN ORDER TO MINIMIZE CUTTING INTO PIPES

GEYSER INSTALLATION TO COMPLY WITH SABS 0254

GROUND STOREY WATER

SCALE 1:50



GROUND STOREY ELECTRICAL

SCALE 1:50

ELECTRICAL NOTES:

1. 3 PHASE CONNECTION TO BE PROVIDED IF POSSIBLE.
2. TELEPHONE ENTRY POINT TO BE PROVIDED AT SUITABLE POINT TO BE DISCUSSED WITH G.P.O.
3. ALL TELEPHONE CONDUITS TO BE SUPPLIED WITH DRAW-WIRE.
4. PROVIDE COVER PLATES TO 100x75 TELEPHONE OUTLETS.
5. SABS APPROVED EARTH LEAKAGE UNIT TO BE PROVIDED.
6. 100x100 T.V. OUTLET BOXES AT 220mm ABOVE FINISHED FLOOR OR AS OTHERWISE INDICATED, LINKED WITH 25mm CONDUIT IN SERIES TO OUTLET BOX ALONGSIDE PLUG OUTLET IN GARAGE.
7. EXTEND CONDUIT TO AERIAL/SATELITE DISH IN POSITION TO BE DISCUSSED WITH ARCHITECT.
8. 21mm SEPARATE EARTH CONDUIT FROM THAT POINT DOWN TO FLOOR AND THROUGH SURFACE BED TO OUTSIDE.
9. THE ELECTRICAL CONTRACTOR MUST CALCULATE THE TOTAL LOAD AND MAKE ALL NECESSARY PROVISIONS FOR THE INSTALLATION.
10. THE REQUIRED AMPS PER PHASE AND CIRCUIT MUST BE CALCULATED AND PROVIDE FOR.
11. APPROVED RATED BREAKERS TO BE PROVIDED FOR EACH CIRCUIT.
12. ALL ELECTRICAL INSTALLATIONS MUST BE SABS APPROVED AND BE DONE BY A REGISTERED AND QUALIFIED ELECTRICIAN.
13. PROPER TESTING TO BE DONE AFTER INSTALLATION AND WIRING CERTIFICATE NEEDS TO BE ISSUED IF ALL IS IN ORDER
14. ALL INSTALLATIONS ARE TO BE AS ENERGY EFFICIENT AS POSSIBLE AND MUST COMPLY WITH SANS 10400XA AND SANS 204
15. ALL LIGHTS TO BE LOW WATTAGE LED OR ENERGY SAVER LIGHT BULBS.
16. PROPER EARTH TO BE INSTALLED AS PER NATIONAL REGULATIONS AND ALL STEEL, INCLUDING WATER PIPES ARE TO BE EARTHED.

SEE RATIONAL DESIGN

NOTE: SOLAR GEYSER SIZE TO BE DETERMINED BY WETWORKS ENGINEER. INSTALLATION TO BE DONE BY A SPECIALIST

SPECIFICATIONS

1. GENERAL
1.1 THIS DRAWING IS NOT TO BE SCALED, USE FIGURED DIMENSIONS ONLY. ALL DIMENSIONS AND HEIGHTS TO BE CHECKED AND VERIFIED BEFORE ANY WORK COMMENCES ON SITE. ANY DISCREPANCIES SHALL BE REPORTED TO THE ARCHITECT IMMEDIATELY. ALL LEVELS, HEIGHTS OF PLINTHS, DEPTH OF EXCAVATIONS AND NUMBER OF STEPS TO BE FINALLY CHECKED BY CONTRACTOR ON SITE.
1.2 THE SITE TO BE TREATED IN ACCORDANCE WITH S.A.B.S. CODE OF PRACTICE NO.0124-1977 WITH 'SHELLDRITE' TERMITEPRUFE SOIL POISONER.
1.3 TOP OF FOUNDATIONS TO BE A MINIMUM OF 300mm BELOW NATURAL GROUND LEVEL.
1.4 BACKFILL TO FOUNDATIONS.
1.5 TOP OF 75mm CONCRETE SURFACE BED TO BE MINIMUM OF 150mm ABOVE FINISHED GROUND LEVEL
1.6 75mm THICK CONCRETE SURFACE BED TO BE ON 'GUNPLAS' USB GREEN DAMP PROOF MEMBRANE ON SAND BINDING ON WELL COMPACTED FILL.
1.7 'GUNDLE' BRICKGRIP S.A.B.S. EMBOSSED D.P.C 375 MIC. UNDER ALL WALLS, WINDOW CILLS AND AT CHANGES IN FLOOR LEVELS.
1.8 STORMWATER SHALL BE REMOVED FROM DWELLING, YARD AND SITE.
1.9 ELECTRICAL INSTALLATION SHALL BE 'HEINEMANN' EARTH LEAKAGE SYSTEM IN DISTRIBUTION BOARD.
1.10 ALL BUILDING WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES BUILDING BY-LAWS AND REGULATIONS.

2. FOUNDATIONS
2.1 ALL FOUNDATIONS, RETAINING WALLS, STRUCTURAL CONCRETE WORK AND SUB SOIL STORMWATER DRAINAGE TO CIVIL ENG'S SPECIFICATIONS.
2.2 ALL SOIL COMPACTION TO CIVIL ENG'S SPECIFICATIONS.
2.3 ALL FOUNDATIONS TO BE REINFORCED AS REQUIRED BY ENGINEER.

3. FLOORS
3.1 MIN. 75mm CONCRETE SURFACE BED.
3.2 25mm CEMENT SCREED TO ALL FLOORS.
3.3 SKIRTINGS TO BE PROVIDED.

4. WALLS
4.1 220mm BRICK WALLS WITH GUNDLE BRICKGRIP DPC.
4.2 BRICKFORCE EVERY 4 COURSES AND EVERY COURSE FOR 4 COURSES OVER OPENINGS
4.3 ALL BRICKWORK MIN. 7 Mpa CLASS II.
4.4 WINDOWS BUILT IN WITH GUNDLE BRICKGRIP DPC.
4.5 ALL PAINTWORK TO COMPLY WITH MANUFACTURER'S SPECIFICATIONS WITH NECESSARY UNDERCOATS.
4.6 EXTERNAL-TO PLAN
INTERNAL-PLASTER AND PAINT.

ALL LEVELS, DIMENSIONS, DEPTHS OF EXCAVATIONS, HEIGHTS OF PLINTHS, NUMBER OF STEPS AND SIZES OF FOUNDATIONS TO BE PLACED AT A HEIGHT OF EXISTING STRUCTURE

5. ROOF A
5.1 CROMADEK SHEETING @ 30° PITCH- DARK DOLPHIN ON 38X50mm S.A. PINE PURLINS @ 1200mm MAX C/C. ON 152X38mm S.A.PINE, GRADE 6, PREFABRICATED TRUSSES @ 1200 C/C ON S.A. PINE 114X38mm WALLPLATES. TRUSSES TO BE TIED DOWN WITH HOOP IRONS THREADED THROUGH FOUR COURSES OF BRICKWORK. PROVIDE SABS APPROVED PLASTIC UNDERLAY AND SISALATION INSULATION.
5.2 MIN.170mm CONCRETE SLAB TO ENG'S DETAIL.
5.3 MIN. 55mm VEMICULITE INSULATING SCREED TO 1:60 FALL.
5.4 WATERPROOFING BY SPECIALIST TO MANUFACTURERS SPEC.
5.5 150Ø P.V.C RAINWATER DOWNPIPES FROM FULLBORE OUTLETS.
5.6 SKYLIGHTS TO MANUFACTURERS SPEC.
5.7 GUTTERS : MARLEY STREAMLINE SQUARE PROFILE RAINWATER GUTTERS + DOWNPIPES.
5.7 3mmSKIM COAT RHINOLITE ON PLASTERBOARD CEILING FIXED TO 38 X 38mm S.A. PINE BRANDERING AT 450 C/C FIXED TO TIE BEAMS.

6. WINDOWS AND DOORS
6.1a ANY PANE OF GLASS WHICH IS TO BE INSTALLED WITHOUT A SUPPORT FRAME SHALL BE IN ACCORDANCE WITH SABS 0137
6.1b THICKNESS OF PANES IN RELATION TO THEIR AREA SHALL BE IN ACCORDANCE WITH SABS 0137
6.2 EXTERNAL SLIDING DOORS : STANDARD AS PER WINDOWS TO SCHEDULE.
6.3 INTERNAL DOORS : STANDARD HARDWOOD FRAMES TO SCHEDULE
6.4 WINDOWS : ALUMINUM TOP-HUNG
6.5 SLIDING DOORS :STANDARD ALUMINIUM SLIDING DOORS TO SCHEDULE.

7. DRAINAGE NOTES
7.1 100Ø uPVC. SEWER PIPE DRAIN WITH A MIN. FALL OF 1:60.
7.2 100Ø OVP. AT HEAD OF DRAIN PIPE AS SHOWN.
7.3 RODDING EYES AT HEAD OF DRAIN, AT ALL CHANGES OF DIRECTION AND AT MAX. OF 25000 INTERVALS.
7.4 INSPECTION EYES AT ALL JUNCTIONS OF DRAIN, AND TO HAVE MARKED COVERS AT GROUND LEVEL.
7.5 DRAIN PIPE UNDER BUILDING TO BE PROTECTED AGAINST LOAD.
7.6 ALL WASTE PIPE TO HAVE 65mm RE-SEAL TRAPS, ALL WASTE PIPES TO BE ACCESSIBLE OVER ENTIRE LENGTH FOR CLEANING AND REPAIRS.
7.7 ALL WASTE PIPES UNDER FLOOR SLAB TO BE SLEEVED.
7.8 ALL SOIL FITTINGS WITH VERTICAL DISCHARGE GREATER THAN 1220 TO HAVE ANTISYPHON VENTPIPES.
7.9 RAINWATER DOWNPIPES TO DISCHARGE A MIN. OF 2440 FROM ANY OPEN GULLY
7.10 ALL DRAINAGE WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES DRAINAGE BY-LAWS AND REGULATIONS.

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PROPOSED STAFF ROOMS
ON PTN 465
OF THE FARM KROMDRAAI 560KQ
CLIENT:

WORKING DRAWINGS

TECHNOLOGISTS: ARCHITECT:	W JUNG-KRUGER
DATE:OCT'18	DWG NO: H/KD
DRAWN: WJK	SHEET : 6 OF 6